Junior Makers · Class 28

Wind Turbine Power Lab

Catch the wind, make electricity, find the best design

Generators | Multimeter | Fair testing | Bar charts



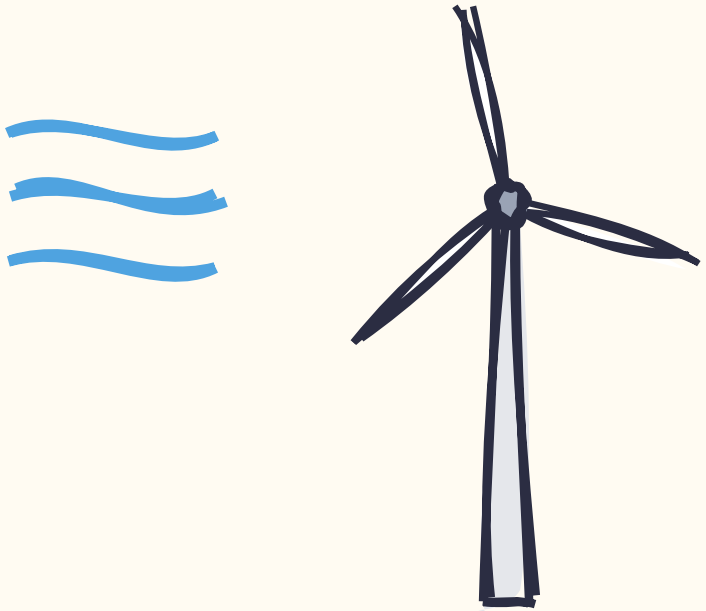
Did you know?
Pakistan's windiest area, the Gharo–Jhimpir corridor in Sindh, has dozens of wind farms making electricity for the grid.

Let's think...

Can you make electricity from wind?

Free energy in the air

Wind is everywhere and costs nothing. Giant turbines turn it into electricity for whole towns. But what is the **best shape** for the blades? How many? How big? Engineers test, and today so will you!



Just add more blades?

Bigger is always better?

Test and find out!

Today: you build a **wind turbine**, measure the electricity it makes, and run **fair tests** to find the best blades.

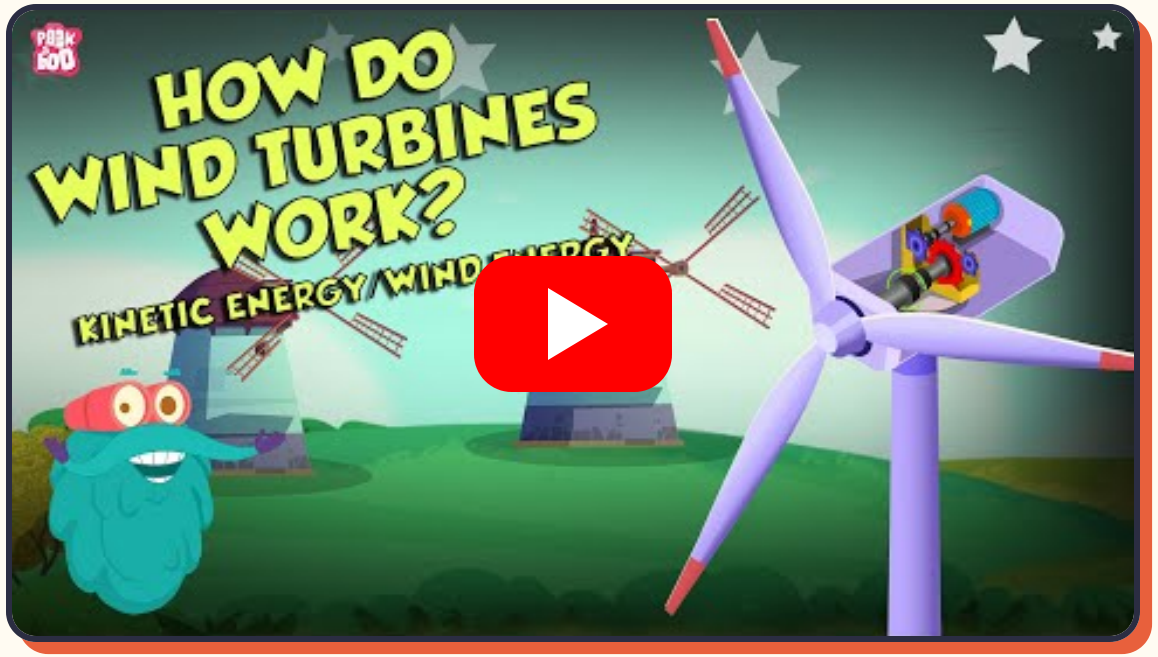
Learning objectives

By the end of this class I can...

1	**Generator:** explain how a turbine turns **moving air into electricity**	**Proof:** explain the energy chain: wind → spin → electricity
2	**Build & measure:** build a turbine and measure its **voltage** with a multimeter	**Proof:** a reading in volts from our turbine
3	**Fair test:** change **one variable** at a time to test blade designs	**Proof:** our table tests 4 designs fairly
4	**Conclude:** use a **bar chart** to pick the best design and explain why	**Proof:** a bar chart and one sentence of evidence

Objective 1 · Generator · video

Video time!



How Do Wind Turbines Work?

The Dr. Binocs Show · about 6 minutes · youtu.be/ia1LJE9sOxQ

Watch for...

1. What makes the blades **spin**?
2. What is inside the box at the top (the **nacelle**)?
3. Why are wind farms built on hills or at sea?

Scan to watch

After the video: Blow on your hand gently, then hard. Which would spin a turbine faster? How could we MEASURE the difference?

Objective 1 · Generator

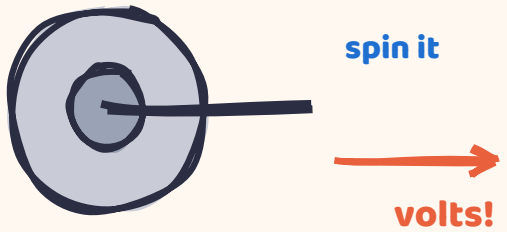
A motor run backwards is a generator

The idea

Inside a DC motor are **magnets** and **coils of wire**. Put electricity in: it spins. Spin it yourself: it makes **electricity**! That's a **generator**.

The energy chain:
moving air → spinning blades → spinning motor → electricity
Faster spin = more voltage.

Example: a bicycle dynamo light



Some bicycles have a little dynamo touching the wheel. Pedal and the light glows; stop and it goes out. Your legs are the 'wind' and the dynamo is the generator.

Objective 3 · Fair test

A fair test: change ONE thing

The idea

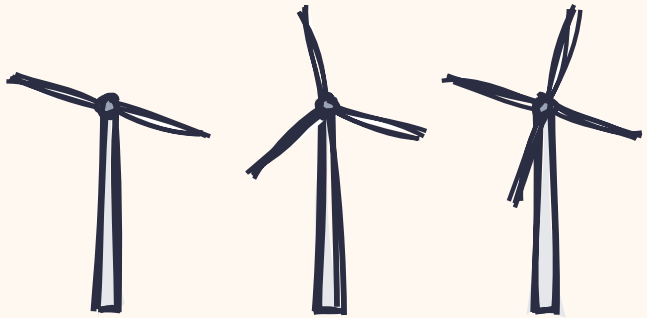
To find out if **number of blades** matters, change only that. Keep everything else the **same**:

- same fan, same distance (30 cm)
- same blade size and angle
- same time to settle

The thing you change is the **variable you test**. The thing you measure (volts) is the **result**.

Repeat
Do each test 3 times and take the average. One lucky reading can fool you!

Example: a running race



To find the fastest runner, everyone must run the same distance on the same track. If one runner gets a shorter track, the race tells you nothing. Same for our turbines.

Predict first!

What do you think will happen?

Before testing, write your prediction

More blades will make **more / less / the same** voltage, because _____.

A steeper angle will make **more / less / the same** voltage, because _____.

Scientists predict

Getting a prediction WRONG is not failing. It's discovering something new!

Objective 4 · Conclude

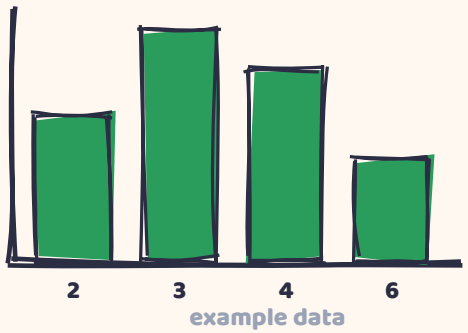
Show results with a bar chart

The idea

Draw one bar per design: **designs along the bottom**, **volts up the side**. The tallest bar is the best design (for our fan).

Then write your conclusion with evidence:
'**3 blades** made the most voltage (**0.48 V** average), more than 6 blades (**0.21 V**).'

Example: a class height chart



If you line up the class from shortest to tallest, you instantly see who is tallest. A bar chart lines up the results so the best design jumps out.

Objective 4 · Conclude

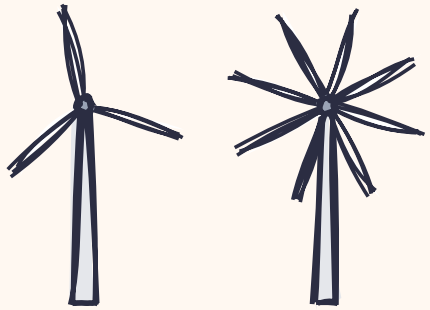
Why isn't 'more blades' always better?

The idea

Each extra blade adds **weight** and **drag** (air resistance). Blades also get in each other's way, stealing the wind.

Real wind turbines usually have **3 blades**: a balance between catching wind, spinning fast, and being steady and cheap to build.

Example: rowing a boat



One rower with long oars can go fast. Twenty rowers in a tiny boat would hit each other's oars and slow down. More isn't better if they get in each other's way.

Real world

Wind power in Pakistan and the world



Jhimpir wind farms
Sindh's windy corridor powers homes across Pakistan



Offshore
Giant turbines at sea where the wind is strongest



Clean energy
No smoke while making electricity



Storage
Batteries save wind energy for still days



Why are wind farms built in Sindh's coastal corridor and not in the middle of a city?

Big blades
Some modern blades are over 100 metres long, longer than a football field!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human energy chain
2. Our turbine kit (per group)
3. Build your turbine
4. Measure the voltage
5. Run the experiments
6. Build your best turbine
7. Bug hunt

Class game

Human energy chain

How to play

1. Line up 5 students: **Sun**, **Wind**, **Blades**, **Generator**, **Light bulb**.
2. Pass a ball down the line: each says how the energy changes. 'Sun heats the air…'
3. Then remove one person. Can the chain still work?
4. Finally: what happens if there is no wind today?

Then discuss

Where does wind's energy really come from? What happens to wind power on a still day?



Big idea

Wind energy comes from the sun heating the Earth unevenly. No wind = no power, so we need batteries or other sources too.

Build it

Our turbine kit (per group)



DC hobby motor
Our generator



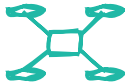
Multimeter
Measures the voltage



Card / chart paper
To cut blades



2 wooden skewers
Hub and supports



5 V fan
Our 'wind' (steady and fair!)



Crocodile clip leads
Motor to multimeter

Also: cork or bottle cap for the hub, tape, ruler, protractor and scissors.

Fan safety
Keep fingers, hair and paper away from the fan's blades. Switch it off before adjusting anything.

Objective 2 · Build step 1 of 3

Build your turbine

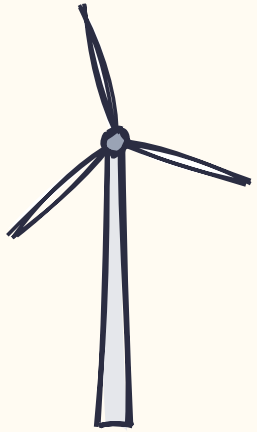
a Push a bottle cap (the hub) onto the motor's shaft. Ask the teacher to help make a tight fit.

b Cut **3 blades** from card, each **8 cm × 3 cm**.

c Tape each blade to a skewer piece and push them into the hub, evenly spaced (120° apart).

d Twist every blade the same small angle (about **20°**) like a fan.

e Tape the motor to a tower (a cup or box) so the blades can spin freely.



Check: Blow on it: the blades spin smoothly without hitting anything.

Blades all flat (0°)? The wind just pushes them and they won't spin. They need a twist!

Objective 2 · Build step 2 of 3

Measure the voltage

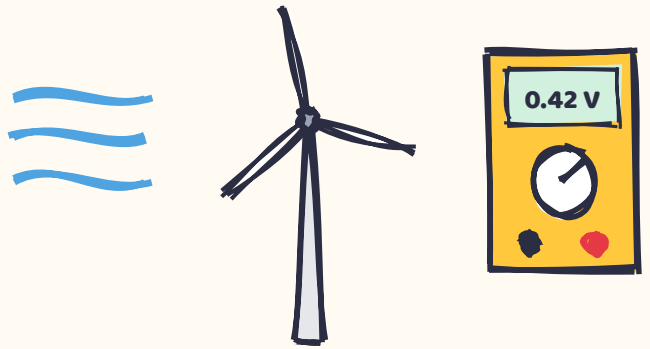
a Set the multimeter to **DC volts** (V with a straight line), the 2 V or 20 V range.

b Clip the motor's two wires to the multimeter's red and black leads.

c Place the fan exactly **30 cm** away, pointing at the blades. Mark the spot with tape!

d Switch the fan on. Wait 5 seconds for the speed to settle.

e Read the voltage. Write it down.



Check: You get a steady reading, usually a few tenths of a volt.

Negative number? The wires are swapped. That's fine, just note it, or swap the clips.

Objective 3 · Fair test · Build step 3 of 3

Run the experiments

Design	Try 1 (V)	Try 2 (V)	Try 3 (V)	Average
2 blades				
3 blades				
4 blades				
6 blades				

Then test

Blade angle: 10°, 20°, 40°

Blade length: 5, 8, 11 cm

(One variable at a time!)

Average

Add the 3 tries, divide by 3.

Objective 4 · Conclude · design challenge

Build your best turbine

Use your evidence

Combine your best blade number, angle and length into ONE design.

Predict its voltage, then test it 3 times. Did it beat all your single tests?

Our best design
Number of blades
Angle
Length
Predicted volts
Measured average

Test like an engineer

Bug hunt

Problem	Likely cause	Did it happen?
Reading is 0.00	multimeter on AC, or wires not clipped	
Blades don't spin	angle 0° or blades rubbing	
Readings jump around	fan distance changing / blades wobbling	
Hub slips on shaft	hole too big: add tape or glue	
Negative voltage	wires swapped (OK!)	

Debug checklist

Tape the fan and turbine positions to the desk so every test is the same.

Small voltages are normal: a hobby motor in a small breeze makes well under 1 volt.

MAKE

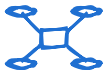
Level up!

Energy challenges

Choose your challenge



Light it: can your turbine light an LED? (Hint: LEDs need about 2 V.)



Wind speed: test at 20, 30, 40 and 50 cm. Graph distance vs volts.



Blade shapes: try curved, triangle and wide blades.



Team power: connect two turbines in series. Does the voltage add up?

Think like an engineer

Why do you think real turbines turn to face the wind?

Boss level

Design a turbine that works even in a gentle breeze (from 60 cm away).

Objective check

Can I really do it? Prove it.

01	Explain the energy chain from wind to electricity.	○ ○ ○
02	Show your multimeter reading. Which setting did you use?	○ ○ ○
03	Which variable did you test? What did you keep the same?	○ ○ ○
04	Show your bar chart. Which design won, and what is your evidence?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Turbine expo: each team presents its best design and its bar chart.

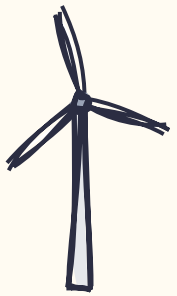
Show what you know

What did we learn?

- [x] A generator turns spinning into electricity
- [x] Wind energy comes from the sun heating the air
- [x] Fair tests change ONE variable and repeat 3 times
- [x] Bar charts show which design is best
- [x] More blades is not always better: drag and weight matter

At home

Find out which power stations supply electricity near you. Is any of it wind or solar?



Next class: Python Number Guessing Game!

Exit ticket: you changed blade number AND angle at the same time. Why is that a problem?